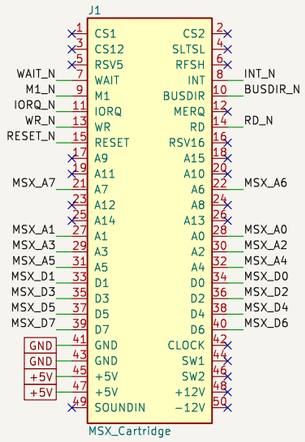
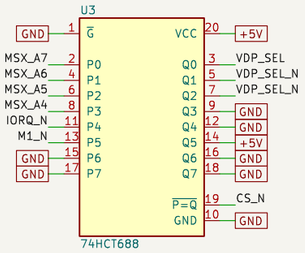
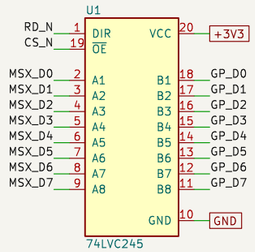
# MSX V9990 (GFX9000) / V9968 cartridge — RP2350 (Waveshare Core2350B)

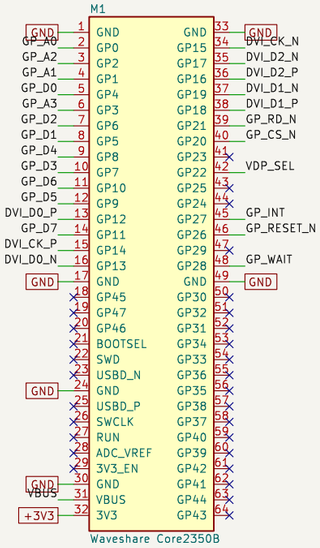
Generated by hardware/gen/gen_schematic.py from design.py — edit the generator, not this file

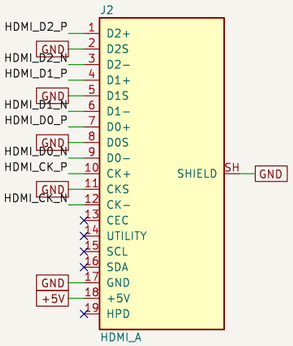
U3: /CS = 0 when /IORQ = 0, /M1 = 1, A7..A4 = 0110 (SW1 V9990: 60h–6Fh) data1000, (SW1=V9968: 98h–9Fh) RP2350 -> MSX), enabled by /CS

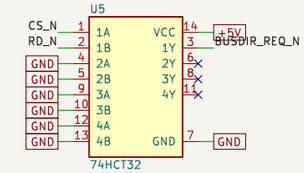


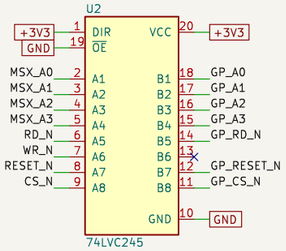



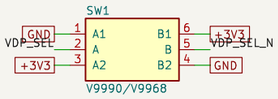



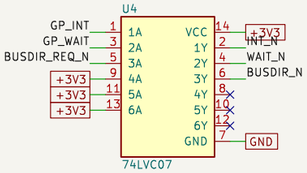



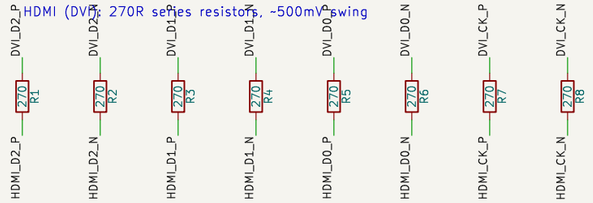

SW1: emulated VDP, read by the firmware on GP22 (VDP_SEL high = V9968)



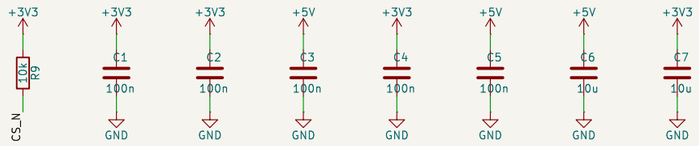

U4: open-drain /INT, /WAIT, /BUSDIR (5V tolerant)

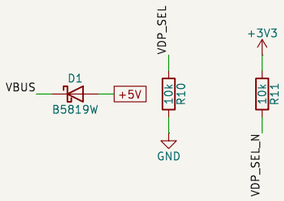



HDMI (DVI): 270R series resistors, ~500mV swing




Decoupling: one 100n per IC, 10u bulk on +5V and +3V3







Core2350B: firmware upload via the module FFC + Waveshare USB adapter (USB, BOOTSEL, RESET).

D1 keeps USB power out of the MSX. Mount the module on pin headers (PSRAM on its back).

Generated from hardware/gen/design.py

Sheet: /
File: msx_v9990_cart_core2350b.kicad_sch

**Title: MSX V9990 / V9968 Cartridge (RP2350, Waveshare Core2350B)**

| Size: A3 | Date: | Rev: 0.2 |
|---|---|---|
| KiCad E.D.A. 8.0.9 | | Id: 1/1 |